



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
(An Institution of National Importance under Ministry of Education, Govt. of India)
MID SEMESTER EXAMINATIONS-I

B. Tech I Semester

MAIC-101: Differential Calculus and Differential Equations

Date: 19-09-2024

Time: 08:30 AM to 09:20AM

Max. Marks: 20

N.B.: Answer ALL questions. Each question carries equal marks.

Answers to all parts of each question should be in one place.

Q1	Give a system of linear equations over field $\mathbb{R}$ (3 equations and 3 unknowns) having a) $(1, 0, 0)$ as the only solution. b) $(1, 0, 0)$ and $(0, 1, 0)$ as solutions. c) $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ as solutions. d) No solution.
Q2	Define linearly independent or linearly dependent vectors. Check whether $U = \{1, x^2, x^3, x^4\}$ is linearly independent or linearly dependent over field $\mathbb{R}$, with proper explanation.
Q3	Write the quadratic form for the given matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Convert that quadratic form into its canonical form over field $\mathbb{R}$. Specify the transformation (modal matrix). Also, specify the orthogonal transformation. Find the rank, index and signature of that quadratic form.
Q4	Identify the given matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ is diagonalizable or not over field $\mathbb{F}$. If matrix A is diagonalizable over a particular field then evaluate $\text{Trace}(B^2 + 3B + 5I)$ and $ B^2 + 3B + 5I $ where $B = A^{10}$.